UI, 2020 CHEMISTRY

1. **The density of a certain gas is 2.5893 g/dm3 at STP. What is the molar mass of the gas? (a.m.v. = 22.4dm3) A. 100g/mol B. 58g/mol C. 44mol D. 32g/mol**

To solve this problem, we need the equation relating density of a gas to its molar mass. Let's get into its derivation!
From general gas equation also called ideal gas equation,
PV = nRT

Recall that, n = m/M

PV = mRT/M

PM = (m/V)RT

Recall that, ρ = m/V

PM = ρRT

That is the equation we need!

M= ρRT/P
Note that at S.T.P,
T = 0^{o}c = 273K , P = 1atm

Note that density given in the option is g/dm3, So we need to use value of R whose unit corresponds to
You can calculate R anytime, at anyday.

From PV = nRT

For standard conditions,
n=1 mole, P = 1atm, V = 22.4dm3, T = 273K
Plugging in the values,
1 atm x 22.4dm3 = (273)R
R = 0.0821atm.dm3/Kmol (This is the value of R we are using since all the units of the components from which it was calculated are consistent with the units we will be dealing with here)

M = (0.0821 x 2.5893 x 273)/ 1
M = 58g/mol.
The correct option is therefore B.

Comment: The value of R to be used depends on the units of temperature, pressure and volume. If pressure is in its SI unit (that is N/m2 or pascal), volume is in m3 and temperature in K, the value of R is 8.314J/Kmol. So, always watch out for units of temperature, pressure and volume before you use value of R.

Significance: You can always calculate the value of R anytime you want to using the units of temperature, pressure and volume whose units corresponds to the problem you are dealing with. Just like what I did here.

Alternative and preferably,
Density of a gas = molar mass of the gas/ molar volume

molar mass = density x molar volume

molar volume = 22.4dm3/mol

molar mass = 2.5893g/dm3 x 22.4dm3/mol

Molar mass = 58g/mol

Using the second method,
Which of the following is the density of nitrogen at STP?
A. 0.33g/L
B. 0.65g/L
C. 0.80g/L
D. 1.25g/L
E. 1.60g/L

Congrats! I believe you did this:
Density of nitrogen gas = molar mass of nitrogen/molar volume
Molar mass of nitrogen gas = 28g/mol

Density = (28g/mol)/22.4L/mol
Density = 1.25g/L

Note: 22.4dm/mol is the same thing as 22.4L/dm3

2. A gas X diffuses twice as fast as gas Y with relative molecular mass of 112, calculate the relative molecular mass of X A. 28 B. 14 C. 56 D. 120 .

Using Graham's law of diffusion of gases.
$R_X/R_Y = \surd RMM_Y/\surd RMM_X$
The question said that gas X diffuses twice as fast as gas Y, this means that rate of X is 2 times the rate of Y
That is $R_X = 2R_Y$

$2RY/RY = \surd 112/ \surd RMM_X$

$2/1 = \surd 112/ \surd RMM_X$

Squaring both sides,

$4/1 = 112/ RMM_X$

RMMX = 28
The correct option is therefore A.
Suppose you are asked the following question in your pending exam, what will your answer be?
When two gases having volumes 1 litres and 2 liters are at the same temperature, then the ratio of the average kinetic energy of the molecules in the two gases
A. 1 : 1
B. 1 : 2
C. 2 : 1
D. 3 : 1

The correct option is A.

The average kinetic is only depends on the absolute temperature. The relation is:
KE=3/2 RT
As you can see, the term 3/2 R is constants. The average kinetic energy only changes when the temperature changes. Since the two gases are at the same temperature, they both therefore have the same kinetic energy. So, the ratio of their kinetic energies is 1:1

3. The Sodium salt of a long chain carboxylic acid possessing cleaning action is (A) A fat (B) A soap (C) A detergent (D) an ester

The correct option B. The sodium salt and potassium salts of long chain carboxylic acid (fatty acid) is known as a soap. A soap is therefore a salt formed from a neutralization reaction between a long chain carboxylic acid and a strong base (NaOH or KOH).

Depending on the nature of alkali used in the production of soap, soap is classified into two.

Hard soap which is the sodium salt of a long chain carboxylic acid
Soft soap which is the potassium salt of a long chain carboxylic acid, and this produces more lather than hard soap.

If you are asked the following question in your forthcoming exam, what will your answer be?
Why is soap referred to as salt?
A. It does not contain hydroxide ions
B. It has water of crystallization
C. It is prepared by reacting fatty acid with a strong base
D. All of the above

The correct option is C.

4. Of the following metals, which one pollutes the air of a big city? (A) Chromium (B) Copper (C) Cadmium (D) Lead

The correct option is D. Lead pollutes the air of a big city.

Lead (Pb) is a metal found naturally in the environment as well as in manufactured products. The major sources of lead emissions have historically been from lead fuels in on-road motor vehicles (such as cars and trucks) and industrial sources.

5. In surgery, metal pins are used for joining together broken bones. These metal pins remain uncorroded in the body. What is the material of these pins? (A) Aluminum (B) Iron (c) Copper (D) Titanium

The correct option is D.

Titanium metal is important for its strength, low density and corrosion resistance. It has better corrosion resistance and less reactivity than aluminium, iron and copper. Hence, Titanium pins are used to join broken bones together as they remain uncorroded and unreactive in the body.

6. Formalin is an aqueous solution of (A) Methyl acetate (B) Propanaone (C) Methanol (D) Ethanol

Formalin is a saturated solution of formaldehyde gas in water. It contains about 40% (by volume) or 37% (by weight) formaldehyde gas along with a small amount of stabilizer. Here, the general stabilizer is 10-12% methanol, and it is useful to prevent the formaldehyde polymerization. Without stabilizer, formaldehyde solution is very unstable, and it tends to polymerize, forming macromolecules that are insoluble.

Methanal should be the correct answer to the question. But it is safe to pick methanol should you don't see methanal in the option as the case here.

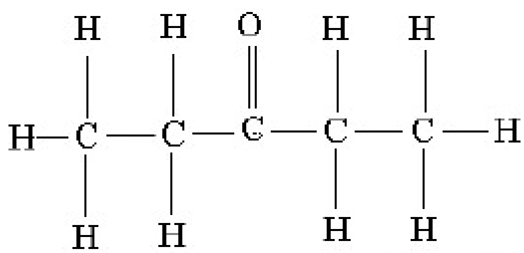
Formaldehyde is used:
A. as an antiseptic and germicide
B. for making synthetic plastics and resins
C. for the preservation of biological specimens
D. all of the above

Correct option is D)

Formaldehyde is used in tanning, preserving, and embalming and as a germicide, fungicide, and insecticide for plants and vegetables, but its largest application is in the production of certain polymeric materials.

7. The IUPAC name of diethyl ketone is (A) propanone (B) Pentan-3-one (C) Butan-2-one (D) Acetone

The correct option is B. Diethyl ketone has the following structure:



Numbering from either side will give the same locant to the carbonyl group. Naming the above compound, we have: Pentan-3-one

8. Which of the following is used as a dehydrating agent? (A) P_2O_5 (B) NaOH (C) dil. HCl (D) $LiAlH_4$

The correct option is A.
Phosphorus(V) oxide is a colorless solid. It has a strong odor. It dissolves in water to produce phosphoric acid (H_3PO_4). It can corrode metals.
Because of its great affinity for water, phosphorus(V) oxide is an excellent drying agent for gases and solvents, and for removing water from many compounds (dehydration action). So, this compound is both dehydrating agent and drying agent. Another compound which is both dehydrating agent and drying agent is conc. H_2SO_4

If the following question is asked in your forthcoming exam, what will your answer be?
A. Conc. H_2SO_4
B. P_2O_5
C. Silica gel
D. All of these

The correct option is D.

9. Which one of the following is produced during the formation of photochemical smog? (A) Hydrocarbons (B) Nitrogen oxide (C) Ozone (D) Methane

The correct option is C.

Photochemical smog is a brown haze is formed when **nitrogen oxides (primary pollutants**) react with **volatile organic compounds (VOCs)**, another primary pollutant in **presence of sunlight**. This photochemical reaction creates secondary pollutants which are the constituent of photochemical smog. When nitrogen dioxide, a primary pollutant, is split by sunlight (UV radiation), this forms nitrogen monoxide and atomic oxygen. The atomic oxygen reacts with oxygen molecule in air to form the **ground-level ozone (also called tropospheric ozone)**. Also, oxides of nitrogen react with hydrocarbons, also known as volatile organic compound (VOC) which is another primary pollutants to produce harmful products like **peroxyacetyl nitrate (PAN).**

As an important note, photochemical smog is a mixture of **primary pollutants (nitrogen oxides and hydrocarbon called volatile organic compounds)** and **secondary pollutants (ozone and PAN)**

Primary pollutants are those pollutants that are released directly from a source. These include oxides of carbon (CO and CO2), sulphur dioxide (SO2) and nitrogen dioxide (NO2).

Secondary pollutants on the other hand are pollutants that are not released directly from a source but are formed by reaction between primary pollutants in air. These pollutants include: Tropospheric ozone (O3), peroxyacetyl nitrate (PAN), nitric acid (HNO3) and sulphuric acid (H2SO4) formed when nitrogen dioxide and sulphur trioxide reacts with water vapour in air respectively.

If the following question comes out in your forthcoming exam, what will your answer be?
Which one causes photochemical smog?
A. O3, PAN and NO2
B. O3, PAN and CO
C. HCN, NO and PAN
D. O2, PAN and NO2

Well done! The answer is A.

Can you attempt this?
A secondary pollutant is:
A. CO
B. CO2
C. PAN
D. Aerosol

Great! The correct option is C.

10. What is the concentration of H+ ion in moles per dm3 of the solution of pH 4.398 A. 4 x 4 x 10-510-2 (B) 4 x 10- 3 (C) 4 x 10- 4 (D) D. 4 x 10- 5

pH = - log [H+]

In this problem, we are asked to find the concentration of the hydrogen concentration, [H+]. So, we have to transpose the formula and extract [H+] out.
Taking negative of both sides,
-pH = log [H+]
Taking antilogarithm of both sides now,
10^{-PH} = [H+]
[H+] = $10^{-4.398}$
[H+] = 3.999 x 10^{-5}
[H+] = 4 x 10^{-5}
The answer is D.

11. When 50cm3 of a saturated solution of a sugar (molar mass 342.0g) at 40^{o}C was evaporated to dryness 34.2g of dry solid was obtained. The solubility of sugar at 40^{o}C is? (A) 2 moles/dm3 (B) 7moles/dm3 (C) 3.5 Moles/dm3 (D) 0.2 moles/dm3

Since 50cm3 of the saturated solution gave 34.2g of the solute on evaporation to dryness. That means, the 50cm3 of the solution essentially contains 34.2g of the solute (sugar).
Let's first find the mass of the solute contained in 1000cm3 (1dm3) of the solution.

50cm3 of the saturated solution ≡ 34.2g of the sugar
1000cm3 of the saturated solution ≡ x
Cross multiply and solve for x
x = 684g
This means that there are 684g of the solute in 1000cm3 of the saturated solution. In other words, the mass concentration is 684g/dm3. And that's the solubility in g/dm3
The next thing is to find the solubility in mol/dm3, bring up the formula relating mass concentration to molar concentration.
molar concentration = mass concentration / molar mass
molar concentration = 684/342
molar concentration = 2 mol/dm3

That's the solubility in mol/dm3. And which option has that? Yes, that's A

12. Zone refining is used for the purification of (A) Au (B) Ge (C) Ag (D) Cu

The correct option is B
We use zone refining to refine semiconductor metals like silicon (Si), germanium (Ge) to obtain them in their purest form. This method effectively removes impurities from semiconducting elements like silicon, germanium, and gallium.
Among the given options for this question, gold (Au), copper (Cu), and silver (Ag) are metals, while germanium (Ge) is a semiconductor. Hence, the correct answer is germanium (Ge).

If the following question is asked in your forthcoming exam, what will your answer be?
Which of the following is used as a method of purification for silicon?
A. Electrolytic refining
B. Liquation
C. Zone refining
D. Distillation

The correct option is C.

13. The iron ore magnetite consists of (A). $Fe_2 O_3$ (B) Fe_3OH_4 (C) $FeCO_3$ (D) $3Fe_2 O_3. 3H_2O$

The correct option is A.
There are two oxide ores of iron – **haematite** with the chemical formula, Fe_2O_3 and **magnetite** with chemical formula of Fe_3O_4.
$FeCO_3$ is a carbonate ore of iron. $FeCO_3$ is called siderite.

Should the following question come out in your imminent exam, what will your answer be?
Siderite is an ore of:

A. Cu
B. Al
C. Ag
D. Fe

Yes! The answer is definitely D.

Kill this!
Haematite is an ore of:
A. Fe
B. Zn
C. Cu
D. Al

Needless to say, the answer is A.

14. The main chemical constituent of clay is (A) silicon oxide (B) aluminum borosilicate (C) Zeolites (D) aluminum silicate

The correct option is B.
Aluminium silicates are minerals composed of aluminium, silicon, and oxygen. They are a major component of kaolin and other clay minerals.

Gun this down!
Which of the following compounds is composed of Al, Si, O and H?
A. Epsom salt
B. Limestone
C. Clay
D. Urea

The correct option is C.
Kaolinite, a type of clay mineral has the chemical formula **Al_2O_3 $2SiO_2 \cdot 2H_2O$**

15. The mineral containing both magnesium and calcium is (A) magnesite (B) calcite (C) carnallite (D) dolomite

The correct option is D.
The chemical formula of dolomite is $MgCO_3$. $CaCO_3$ which can also be written as: $CaMg(CO_3)_2$
Note, magnesite (not magnetite) is a carbonate ore of magnesium only. The chemical formula is $MgCO_3$
Calcite is a carbonate ore of calcium only. The chemical formula is therefore $CaCO_3$.
Carnalite is an halide ore of both potassium and magnesium. Carnalite is $KCl.MgCl_2.6H_2O$

If the following question, comes out in your forthcoming exam, what will your answer be?
Which of the following is a halide ore?

A. Cassiterite
B. Anglesite
C. Siderite
D. Carnallite

The correct option is D.
Cassiterite also called tin stone is an oxide ore of tin. The formula is SnO_2
Anglesite is a sulphate ore of lead. Anglesite is $PbSO_4$
Siderite is $FeCO_3$

16. The metal does not give H_2 on treatment with dilute HCL is (A) Zn (B) Fe (C) Ag (D) Ca

The correct option is C.

An element lower in an electrochemical series cannot displace an element above it from its compounds. Silver being lower than hydrogen in an electrochemical series cannot displace hydrogen from HCl to liberate hydrogen gas.

What if the question comes in another structure like the following, what will your answer be?
Which does not react with dilute hydrochloric acid?

A. Copper
B. Iron
C. Zinc
D. Tin
Congrats! The correct option is A!

What will your answer be?
Which of the following will give H_2 gas with dilute HNO_3?
A. Mg
B. Zn
C. Cu
D. Hg

The correct option is A. Metals react with nitric acid, but hydrogen gas is not evolved, because nitric acid is a strong oxidising agent. So, it oxidises the hydrogen to water and itself gets reduced to any nitrogen oxide. But, Magnesium and Manganese are two metals that traditionally, liberate H_2 and not H_2O when they react with HNO_3

17. **The maximum number of isomers for an alkane with molecular formula C_4H_8 is (A) 5 (B) 4 (C) 2 (D) 3**

Using this formula as a cheating code

Number of isomers = $2^{n-4} + 1$
Our n = 4

Number of isomers = $2^{4-4} + 1$
Number of isomers = 2

As an important note, this formula is valid within n=2 and n=7
When n=8, the formula is adjusted to $2^{n-4} + 2$
When n=9, the formula is adjusted to $2^{n-4} + 3$

If the following question is asked in your pending exam, what will your answer be?
How many isomers C_5H_{12} can have?
A. 3
B. 2
C. 4
D. 5

n is still within the validity of the first formula,
Using the formula, we have
Number of isomers = $2^{5-4} + 1$
Number of isomers = 3
So, the correct option is A.

Gun this down!
Number of chain isomers shown by nonane are:
A. 5
B. 9
C. 18
D. 35

The correct option is D.
The formula to use is: number of isomers = $2^{n-4} + 3$

18. The isomerism which exists between CH_3CHCl_2 and CH_2ClCH_2Cl is (A) chain isomerism (B) functional group isomerism (C) positional isomerism (D) metamerism

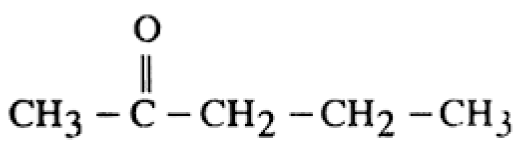
Naming both compounds, CH_3CHCl_2 is 1,1-dichloroethane while CH_2ClCH_2Cl's name is 1,2-dichloroethane.
The two isomers are not chain or skeletal isomers since the length of their carbon chain is the same. So, they cannot exhibit chain isomerism
The two isomers have the same functional group (chloro atoms), so they are not functional group isomers.
The correct option is C. They are positional isomers because the position of the functional group is different. In CH_3CHCl_2, both chloro atoms are on the position 1, while in CH_2ClCH_2Cl, one chloro atom is

on position 1 while the other is on position 2. Got the gist? Great!

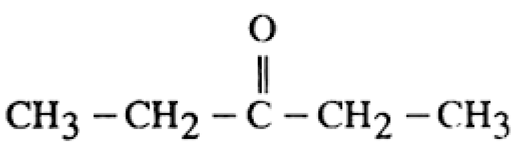
What will you choose if the following question comes out in your exam in 1 day or 2 days away?
Which of the following pairs is an example of position isomerism?
A. CH_3-CH_2-CH_2-CH_3 and CH_3 – $CH(CH_3)$ – CH_3
B. CH_3 – CH_2 – CH = CH_2 and CH_3 – CH = CH – CH_3
C. CH_3 – CH_2OH and CH_2 – O – CH_3
D. CH_3 – $C(CH_3)_2$ – CH_3 and CH_3 – CH_2 – CH_2 – CH_2 – CH_3

Congrats! The answer is B. As you can see, the position of the double bond (a functional group of alkene) is different in both. Note, pair is C are functional group isomers.

Let’s talk a little bit on metamerism. Metamerism is a type of structural isomerism in which the isomers called metamers have the same molecular formula but different alkyl groups around the functional group. This phenomenon is shown by ethers, esters, ketones, amines, amides etc.



Pentan-2-one



Pentan-3-one

Both have the same molecular formula. Check that out! Have you? Now, take a look at the alkyl groups surrounding the carbonyl group (C=O), what is your observation? Yes, the alkyl groups surrounding the carbonyl group are different. This is what is called metamerism. Let’s take a look at another structures that show this.

$CH_3-CH_2-NH-CH_2-CH_3$

N-Ethylethanamine

$CH_3-NH-CH_2-CH_2-CH_3$

N-methyl-1-propanamine

Go on and answer the following question.

Which of the following compounds exhibit metamerism?

A. Alkanes

B. Alcohols

C. Both A and B.

D. Ethers.

The answer is D.

19. The metal that is usually extracted from sea water is (A) Ca (B) Na (C) K (D) Mg

The correct option is D.

Magnesium is obtained principally with the Dow process (not Down process used for sodium), by electrolysis of fused magnesium chloride from brine and sea water

Magnesium can also be extracted using Pidgeon process. Pidgeon process is one of the methods of magnesium metal production via a silicothermic reaction.

$Si(s)+2MgO(s) \rightarrow SiO2(s)+2Mg(g)$

If you are asked the following question in your forthcoming exam, what will your answer be?

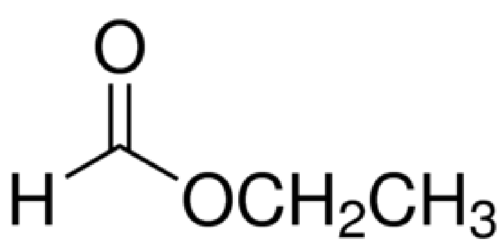
The metal obtained in the Pidgeon process is:

A. Na

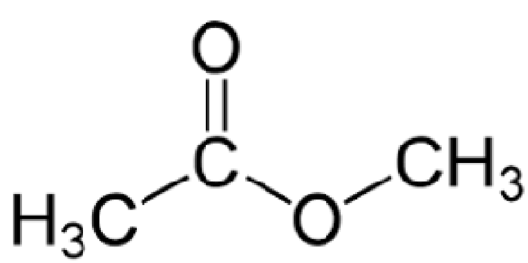
B. Mg

C. Ca

D. Al



Ethyl formate



Methyl acetate

To further prepare you,

What will you choose?

Mond process is used in the extraction of :

The answer is Nickel!
20. An organic compound has the empirical formula CH_2O. What is its molecular formula if its vapour density is 90? (A) C4H304 (B) C12H3201 (C) C4H1005 (D) C6H1206.

The relationship between empirical formula and molecular formula is:
(E.F)n = M.F
Vapour density of a gas = relative molecular mass of the gas / relative molecular mass of H_2

Since relative molecular mass of hydrogen molecule is 2,
Vapour density of a gas = relative molecular mass of the gas /2

So the vapour density of a gas is equal to the half of its relative molecular mass.
In other words, the relative molecular mass of a gas is twice its vapour density.

Relative molecular mass of the compound is therefore, 2 x 90 = 180

$(CH_2O)n = 180$
Summing up the relative atomic masses of the atoms in the parenthesis, we have
30n = 180
n = 6
Now, replace n with the value gotten in the relation between E.F and M.F
MF = $(CH_2O)_6$
MF = $C_6H_{12}O_6$

The correct option is therefore D.

Give this a shot!
For a gas, the relative molecular mass is equal to 2Y. What is Y?

A. The mas of the gas
B. The vapour density of the gas
C. The volume of the gas
D. The temperature of the gas

The correct option is B.

21. What mass of a divalent metal, M (atomic mass 40) would react with excess hydrochloric acid to liberate 224cm3 of dry hydrogen gas measured at stp.? (G.M.V. = 22.4dm3) (A) 8g (B) 0.04 (C) 10g

The reaction equation is M + HCl ----> ?

Since it is a divalent metal, the formula of the compound it will produce when is reacted with HCl is $M(Cl)_2$.

The balanced reaction equation is hence:

$M(s) + 2HCl(aq) \longrightarrow M(Cl)_2(aq) + H_2(g)$

Using mass - volume relationship.

From the reaction equation,

40g of M effervesces $22.4dm^3$ (that is $22400cm^3$) of hydrogen

xg of M would effervesce $224cm^3$ of hydrogen.

Cross- multiply and solve for x

x = (40 x 224)/22400

x = 0.4g

Now, the answer is 0.4g but 0.04g is in the option, you are advised to go for the closest option which is 0.04g.

22. 50cm3 of hydrogen is sparked with 20cm3 of oxygen at 100^0C. The total volume of residual gases is? (A) 50cm3 (B) 10cm3 (C) 40cm3 (D) 30cm3

The equation of the reaction is as follows:
$2H_2(g) + O_2(g) \longrightarrow 2H_2O(g)$
Since we are dealing with gases, Gay-Lussac's law of combining volumes suits here.

$2H_2(g) + O_2(g) \longrightarrow 2H_2O(g)$
2v 1v 2v

From the volume ratio, 2 volumes of hydrogen need 1 volume of oxygen for complete combustion.
In the question, we are provided with both volumes of the reactants.
Whenever a question provides you with amounts of both reactants in a reaction, you need to do a check-up to find out if one is in excess of the other.

We can use either of the volumes given to do the investigation. Let's use that of hydrogen
From the equation,
2 volume of hydrogen = 1 volume of oxygen
50cm3 of hydrogen = x volume of oxygen

Cross-multiply and solve for x
x = 25cm3
What this means is that we would need 25cm3 of oxygen for the complete combustion of 50cm3 of hydrogen given. But we don't have up to this amount as you can see from the question. We only have 20cm3 which is less than the amount that would be required. So, oxygen is said to be in shortage! And hence it is the limiting reagent. Therefore, the volume we will be using to calculate other volumes is oxygen's.

To calculate the volume of hydrogen that reacted, we use
1 volume of oxygen = 2 volumes of oxygen (this is from the balanced equation)
20cm3 of oxygen = x
Solving for x
x = 40cm3

40cm3 of hydrogen out of 50cm3 provided reacted with oxygen to form steam. The remaining 10cm3 will remain unreacted.

Let's calculate the volume of steam formed.
1 volume of hydrogen = 2 volume of steam
20cm3 of oxygen = x
Solve for x, we have
x = 40cm3

So, in the reaction mixture, you will have steam and unreacted hydrogen at the end of the reaction.
volume of the residual gas = volume of the unreacted gas + volume of the steam produced.
volume of the residual gas = 10 + 40 = 50cm3

So, the correct option is A.

Comment:

23. The molality of 2% by weight of aqueous solution of H2SO4 (molecular weight of 98) (A) 0.0051 (B) 0.00051 (C) 0.051 (D) 0.51

Molality is given by the formula:

Molality (m) = number of moles/ mass of the solvent (in kg)

The interpretation of 2% by weight of H2SO4 means that there are 2g of H2SO4 in 100g of aqueous solution of H2SO4.

Number of moles = mass /molar mass

Number of moles = 2/98

Number of moles of H2SO4 = 0.0204

Note that the mass of the solution = 100g.

Solution = solute + solvent

Mass of the solvent = 100 - 2 = 98g

Converting that to kg

Mass of the solvent = 98/1000 = 0.098kg

Molality = 0.0204/0.098

Molality = 0.2m

There is no correct option.

If you are asked the following in your pending exam, what will your answer be?
Molarity is defined as:
A. the number of moles of solute dissolved in one dm^3 of the solution
B. the number of moles of solute dissolved in 1kg of solvent
C. the number of moles of solute dissolved in $1dm^3$ of the solvent
D. the number of moles of solute dissolved in 100ml of the solvent.

The correct option is A.
Note that definition in option B is for molality

24. If $25cm^3$ of $0.160mol/dm^3$ KOH is added to $50cm^3$ of $0.100mol/dm^3$ HNO_3, what is the pH of the resulting solution (A) 7 (B) 1 (C) 2.10 (D) 1.88

When you see an acid base reaction calculation problem where the concentrations and volumes of both the acid and the base are given, the question is to instance incomplete neutralization. In that scenario, the acid and the base will not be present in their equivalents (that is in their right amounts).

The acid might be present in excess of the base and this will make the solution to be acidic or the base might be in excess of the acid and in that case the solution will be basic.

So what we need to do is to first of all do some checkups.

First thing we need to do is to write the equation of the reaction.

The equation of the reaction is:

$KOH + HNO_3 \longrightarrow KNO_3 + H_2O$

As you can see, 1 mole of KOH requires 1 mole of HNO_3 to completely neutralize themselves.

Let's calculate the reacting number of moles in the question and give our judgement.

Number of moles = molar conc x volume in dm^3

Number of moles of KOH = 0.160 x 25/1000

$=4 \times 10^{-3}$mol

Number of moles of HNO_3 = 0.100 x 50/1000

$= 5 \times 10^{-3}$
Now, as you can see, the reacting number of moles of HNO_3 (the acid) is greater than that of KOH (the base). So the resulting solution will be acidic (since the acid in the excess).

Next, we calculate the amount by which the acid in the excess of the base.

$5 \times 10^{-3} - 4 \times 10^{-3} = 1 \times 10^{-3}$.

So the acid is in excess by 0.001mol and this is what caused the acidity of the solution.

Normally, complete neutralization of KOH (a strong base) and HNO3 (a strong acid) is supposed to produced a neutral solution but based on the excessiveness of the acid, the solution is acidic.

Now, what we want to do next is to calculate the molarity of this excess acid

Note, molarity = number of moles /volume of solution in dm^3.

The volume of the solution of mixing the acid with the base = volume of the acid + volume of the base

Volume of the solution = 25 + 50 = $75cm^3$, in dm^3, that gives us,

$0.075dm^3$

Molarity of the excess acid = 0.001/0. 075

Molarity = 0.013M
Now, we want to calculate the pH of the solution imparted by this excess acid.

Note that HNO3 is a strong acid, so it's dissociation is complete (that's 100%). So, 0.013M of HNO3 will release 0.013M of H+

pH = -log [H+]

pH = -log (0.013)

pH = 1.886

That's option D!

It's likely that the following question come out.

A solution is prepared by adding 50.0mL of 0.5M HCl to 150.0mL of 0.10M HNO_3. Calculate the concentrations of all the species in this solution, and the pH of the solution.

Since both acids are strong acids, we can use the following shortcut to get the total [H+] of the resulting solution.

(Volume of HCl) x (concentration of HCl) + Volume of HNO_3 x (concentration of HNO_3) = Total volume of the resulting solution x concentration of the resulting solution.

Using symbols,
$C_1V_1 + C_2V_2 = C_3V_3$
$0.5 \times 50 + 0.10 \times 150 = C_3 (200)$
$C_3 = 40/200$
$C_3 = 0.2M$

pH = -log [H+]
pH = -log 0.2
pH = 0.699

Comment: You would notice that I didn't convert the volumes in unit of mL to unit of L or dm^3. This is because, the unit will cancel out at the end.

Note: The volume of the resulting solution, V_3 is gotten by adding the individual volumes of the two solutions mixed.

Let's take a look at another problem.

The pH **of a solution is** 10 **and that of another is** 12**. When equal volumes of these two are mixed, the** pH **of the resulting solution is:**

A.) 10
B.) 12
C.) 11
D.) 10.30

Note:
Always remember that when equal volumes of two strong acid solutions are mixed together then the resultant is not the average of the pH values of both solutions but rather it is the average of the concentration of hydrogen ion that is given as [H+].

pH=−log[H+]
Or we can also write it as,
$[H+]=10^{-pH}$
Now, for the first solution we have given the pH=10, therefore the value of concentration of hydrogen ion for solution 1 is $[H+]=10^{-10}$
Also, for the second solution we have given the pH=12, therefore the value of concentration of hydrogen ion for solution 2 is $[H+]=10^{-12}$
As we know that when the equal volume of two solutions are mixed together then the concentration of [H+] for each solution gets half. Therefore, the concentration of hydrogen ion for final solution is given as:

$[H+]=[H+]_1/2 +[H+]_2/2$

$[H+]=10^{-10}/2 + 10^{-12}/2$

$[H+]=10^{-10} \times (0.5+0.005)$ (If you undermaths, you we know how we arrived here)

$[H+]=0.505\times10^{-10}$

Now, pH of the resulting solution is :

$pH=-\log[0.505\times10-10]$
$pH=-\{\log[0.505]+\log[10-10]\}$
$pH=-\{-0.30-10\}$
$pH=10.30$

As the required pH value of the resultant solution is approximately 10.3010.30.

Hence, option D.) is the correct answer.

25. In the reaction $SO_2 + \frac{1}{2} O_2$ ----> SO3 ΔG(100K) = -5.19kJ What is the equilibrium constant Kp for the reaction at 100k (R = 8.314J/mol.k.) (A) -1.87 (B) 0.27 (C) -453 (D) 1.87

The relationship between Gibb free energy and reaction quotient, Q is:
$\Delta G = \Delta G^o + 2.303RT \log Q$
At equilibrium, $\Delta G = 0$, $Q = K$

$0 = \Delta G^o + 2.303RT \log K$

$\Delta G^o = -2.303RT \log K$
We can replace K with Kp.
$\Delta G^o = -2.303RT \log Kp$
As you should know, ΔG^o represents standard Gibb free energy change. The temperature of standard Gibb free energy change is 25^{o}c (that is 298K). And as you can see, Gibb free energy we were given is at 100K. Don't fret!
Gibb free energy change does not change appreciably with temperature, so we can use ΔG at 100K for ΔG at 298K

$-5.19 \times 1000 = -2.303 \times 8.314 \times 100 \log Kp$
$2.71 = \log Kp$
Taking antilogarithm of both sides,
$Kp = 10^{2.71}$
$Kp = 512.86$

The question has flaws.

Courtesy: Dr. Adura